High-Frequency Class-D Audio Amplifier

with Digital Feedback Compensation

ENGR489 Progress Presentation | 2026

200 kHz switching · 20 Hz – 20 kHz audio · 8 Ω load · Half-bridge topology · STM32G474RE

Why Is Class-D Hard to Get Right?

Class-AB (Linear)

Low distortion
Poor efficiency ~50%
Simple feedback

VS

Class-D (Switching)

>90% efficiency
But: nonlinear switching
Feedback is complicated

The Core Tension

- Switching nonlinearity → THD
- Supply noise coupling → poor PSRR
- High-speed feedback → stability and EMI risk

How do you close a feedback loop at 200 kHz without introducing more distortion than you remove?

What the Literature Tells Us — and Where It Falls Short

01

Carrier Nonlinearity → THD

RC-generated exponential carrier introduces harmonic distortion. Nonlinearity parameter x must stay between 1.5τ–2.0τ.

02

Feedback Helps, But Adds Stability Limits

Closed-loop control improves THD, PSRR, and switching-error correction, but the loop must be carefully designed around PWM delay, LC resonance, and phase margin limits.

03

PS-IMD Often Exceeds PSRR Distortion

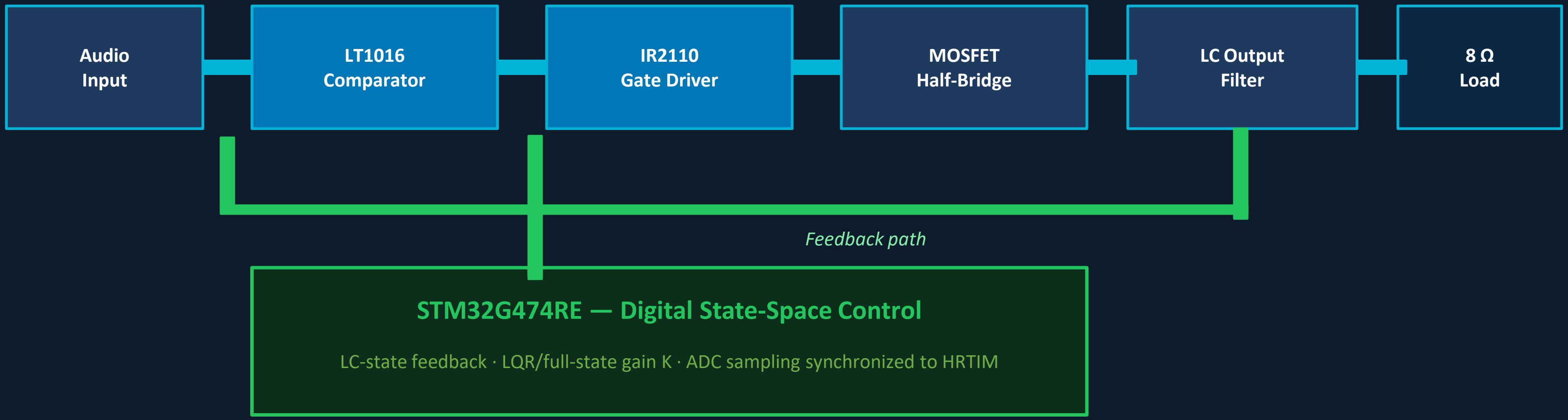
Supply noise intermodulates with the audio signal. PS-IMD can be >20 dB larger than the direct supply distortion.

04

LQR Control: 30× THD Reduction

State-space + full-state feedback (LQR) at high switching frequencies gives dramatic THD+N improvement.

Proposed Closed-Loop Class-D System



Switching Freq	Audio BW	Load	Topology
200 kHz	**20 Hz – 20 kHz**	**8 Ω**	**Half-Bridge**

Three Critical Choices — and Why

1

Half-Bridge MOSFET Power Stage

Selected because it is simpler to implement, requires fewer MOSFETs and gate-drive channels, and is more realistic for an honors-level prototype.

⚠ *Trade-off: alf-bridge designs have poorer supply-noise rejection than full-bridge designs, so feedback control and careful supply decoupling become more important.*

2

Switching Frequency: 200 kHz

Chosen to keep the PWM carrier above the audio band while remaining achievable with the selected gate driver, MOSFETs, and STM32 timing hardware.

⚠ *Trade-off: Higher switching frequency reduces output-filter size and pushes switching noise away from audio, but increases switching losses, EMI, and timing sensitivity.*

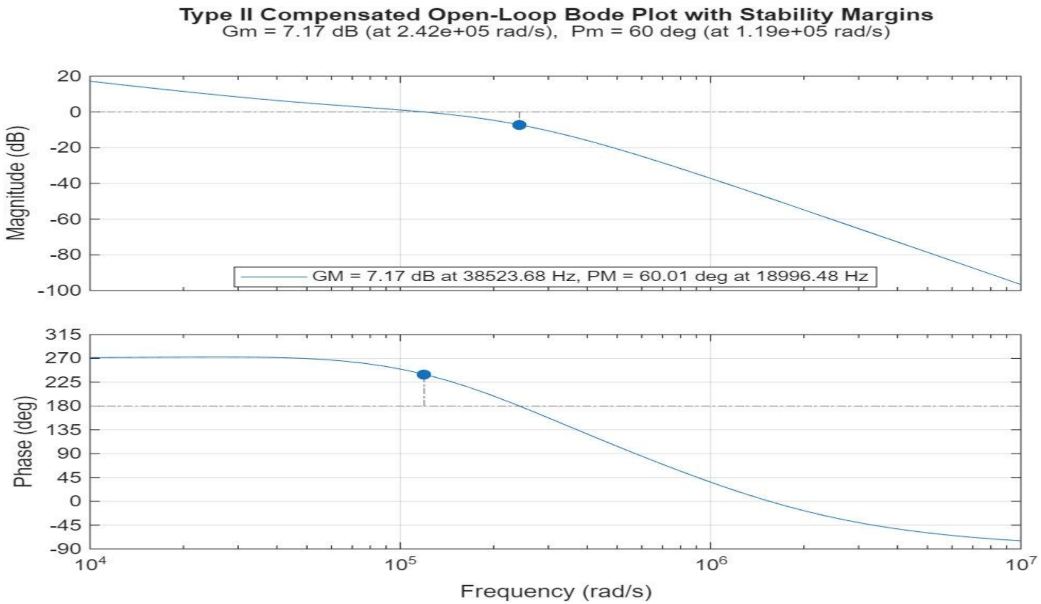
3

STM32G474RE

HRTIM generates the PWM directly with 184 ps resolution, while the FMAC runs the compensation filter with zero CPU overhead.

⚠ *Trade-off:* Introduces computational delay and sampling quantization — both absent in analog controllers — requiring careful phase margin budgeting and anti-alias filtering.

Stability Analysis & Filter Output



Type II Compensated Bode Plot — MATLAB

Phase Margin

60°

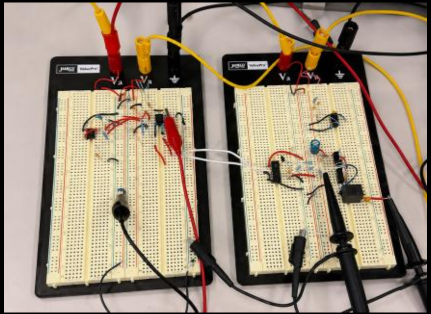
Gain Margin

7.17 dB

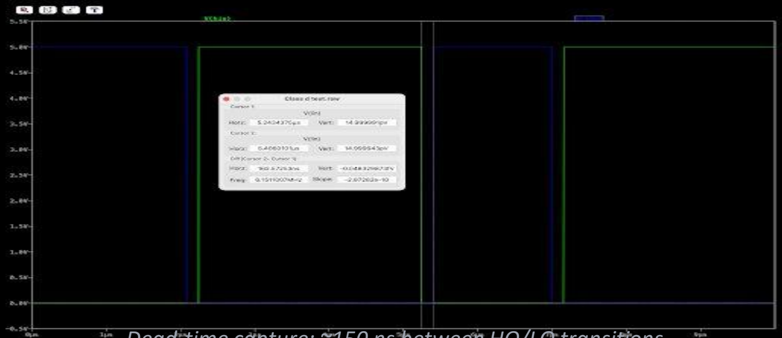
✓ Stable — meets design target
PM > 45° confirms adequate phase reserve at 260 kHz operation

LTspice LC Filter Output — Compensated open-loop

What's Been Built & Tested



Breadboard circuit: Class D audio amplifier circuit



Dead-time capture: ~150 ns between HO/LO transitions

✓ Completed

- IR2110 bootstrap circuit verified — switching cleanly at 200 kHz
- Half-bridge gate drive waveforms confirmed, dead-time ~150 ns
- LC output filter characterised in LTspice
- Type-II compensator designed and stability verified in MATLAB

⌛ In Progress

- Closed-loop hardware integration & bench stability test
- STM32 digital compensator firmware (HRTIM + FMAC IIR)
- THD measurement — LTspice FFT (simulation)
- THD measurement — spectrum analyzer (hardware)

What We're Solving, How We'll Validate, and What's Next

Problem: Insufficient gain margin at 200kHz

→ *Increase switching to 250kHz, increases the Am & Φm bandwidth*

Problem: 2^{nd} filter inadequate for high frequency filtering

→ *Implement a 4^{th} order Butterworth filter for more aggressive attenuation*

Problem: Breadboard layout introduces parasitic inductance, ground bounce, and switching noise

→ PCB implementation to minimize THD through improved switching integrity, cleaner feedback measurements, and reduced EMI coupling

Success criteria: THD < 0.01% closed-loop across 20 Hz–20 kHz

Remaining Timeline

Weeks 14–16

Close the hardware loop; verify stability on bench

Weeks 17–18

THD measurement 20 Hz–20 kHz, open vs closed

Weeks 19–23

STM32 compensator tuned; PSRR improvement measured

Weeks 23 - 24

Final characterisation & report writing

THD Measurement Method

LTspice FFT (simulation) → Spectrum analyzer on hardware
Hann window · 8192 pts · 100Hz / 1kHz / 6.6kHz · 8Ω resistive load

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Half-Bridge Class-D Audio Amplifier

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THD, IMD, PSRR

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LQR and Application for THD Reduction

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